

# Agentic AI for Oracle AI Database

## 1) Oracle AI Vector Search

- \* Builds vector powered applications.

## 2) Oracle AI Database Private Agent Factory

- \* Creates enterprise AI agents without writing the code.

## 3) Select AI Agent

- \* Builds autonomous agents that live inside the Oracle AI Database.

## 4) Oracle Autonomous AI Database MCP Server

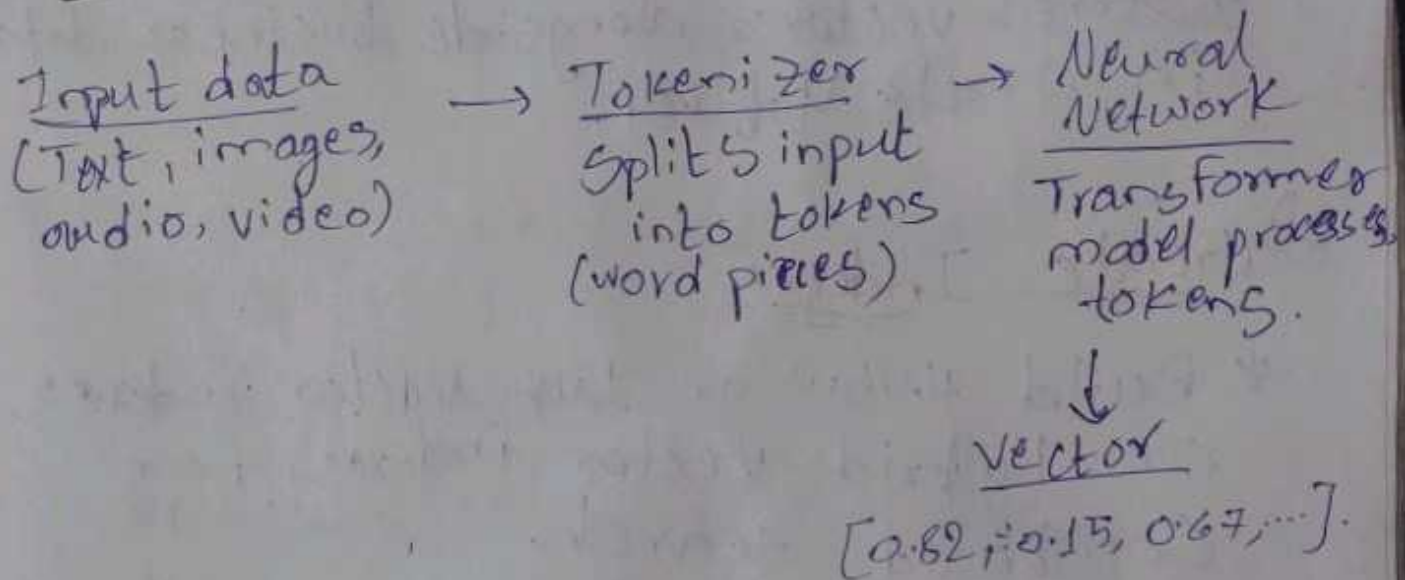
- \* Leverages MCP server built-in the database.

## Vectors & Embeddings

Vectors  $\Rightarrow$  A vector is simply an ordered list of numbers that represents the "location" of something in a high-dimensional space.

Embeddings  $\Rightarrow$  It is a vector created by a neural network that captures the semantic meaning of input data.

## How Embeddings are created



## Oracle AI Vector Search

- \* It is a feature built directly into Oracle AI Database that lets you store, index, & search vector embeddings alongside the traditional business data - all in one system using SQL.

### Workflow

#### 1.) Generate Vector Embeddings

- \* Generate embeddings inside or outside Oracle Database.

##### In-Database

- \* This is done through ONNX Runtime.
- \* Uses ONNX models & generates embeddings using SQL.

##### External

- \* Done through REST API's.
- \* Calls third party API's.

## 2) Store vectors

\* Store vectors alongside business data in Oracle tables.

## 3) Create Indexes

\* Build HNSW or IVF vector indexes, or Hybrid vector indexes for combined search.

## 4) Search & Query

\* Run similarity searches with VECTOR\_DISTANCE(), combine with relational WHERE clause.

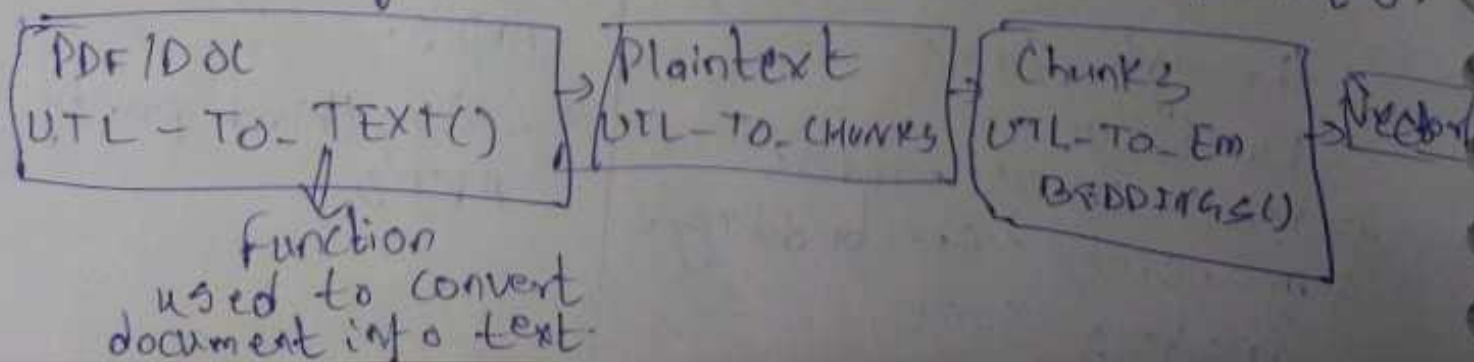
## 5) RAG & LLM

Feed search results into LLMs for RAG responses.

## Chunking

\* Splitting documents for Embedding.

\* Embedding models have token limits.



# Distance Functions & Similarity Search

Finding similar = measuring distance  
vectors between them.

\* Use the same distance metric throughout the process.

## 1.) COSINE (COSINE-DISTANCE)

\* Measures the angle between vectors.

## 2.) Euclidean (L2-DISTANCE())

\* Measures straight line distance between points.

## 3.) Dot Product (INNER-PRODUCT())

\* Measures both magnitude & direction.

## 4.) MANHATTAN (L1-DISTANCE())

\* Sum of absolute differences along each dimension.

## Autonomous AI Database MCP Server

\* A multi-tenant, built-in feature of Autonomous AI Database that exposes Model Context Protocol (MCP) endpoints.

\* Runs inside the database.

\* Skips schema discovery

\* DB enforces every policy.

MCP Client



HTTPS endpoint



Autonomous <sup>AI</sup> User DB



select AI Agent tools.

## Oracle AI Database Private Agent Factory

\* This is a no-code platform that lets business users & AI engineers rapidly build, test, & deploy intelligent AI agents, without writing any code.